INDIAN ECONOMY

LEVEL II




LECTURES

AN OVERVIEW OF THE AGRICULTURE AND ALLIED SECTORS IN INDIA – PART II

MODULE - 52

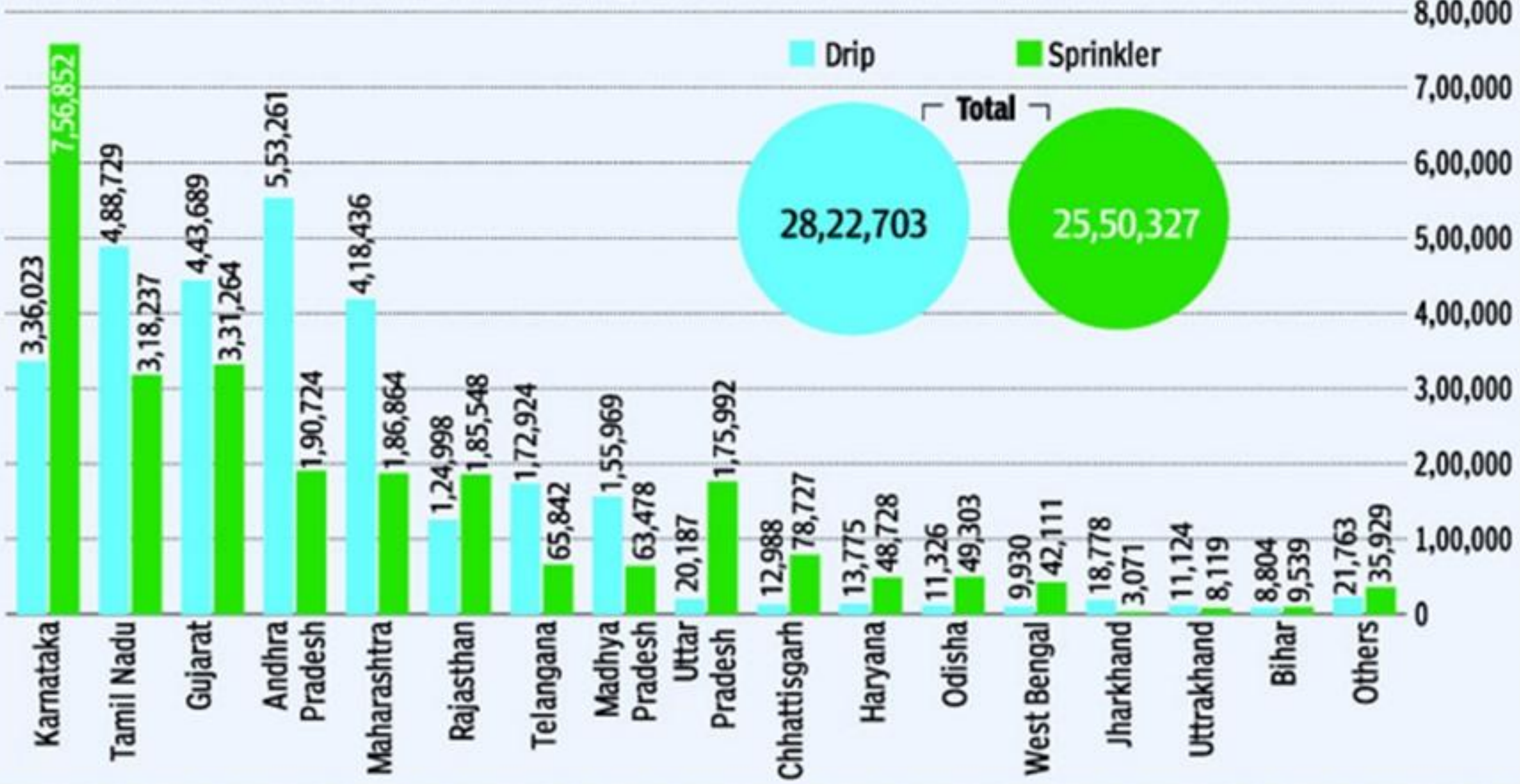
MICRO IRRIGATION UNDER PMKSY … SOUTHERN AND WESTERN STATES TOOK THE LEAD!

State-wise coverage of micro-irrigation and micro-level water harvesting/ secondary storage structures

Created under PMKSY-PDMC as on 2.3.2021

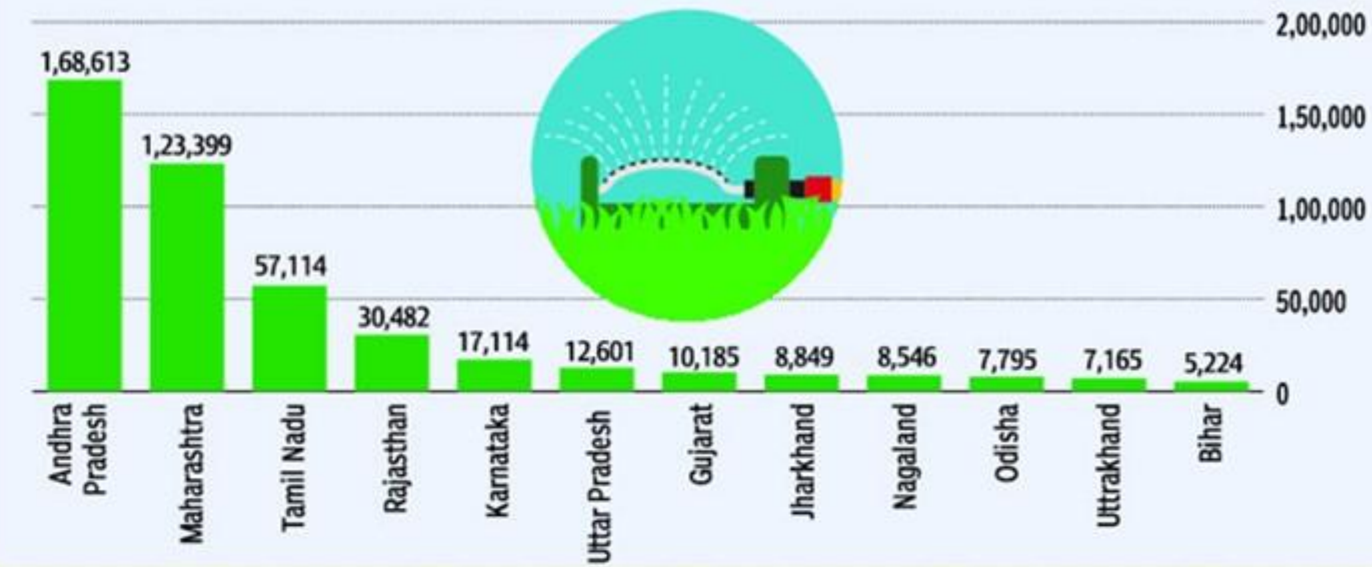
The top States in drip and sprinkler irrigation

(Area in hectares)



MICRO LEVEL WATER HARVESTING STRUCTURES ... SOUTHERN AND WESTERN STATES AGAIN LEAD!

States with most micro-level water harvesting/secondary storage structures



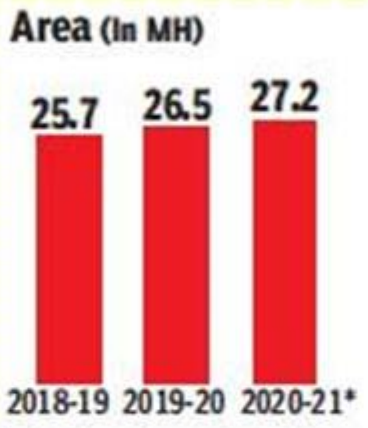
LEVEL OF MECHANISATION IS HIGH FOR WHEAT IN COMPARISON TO RICE!

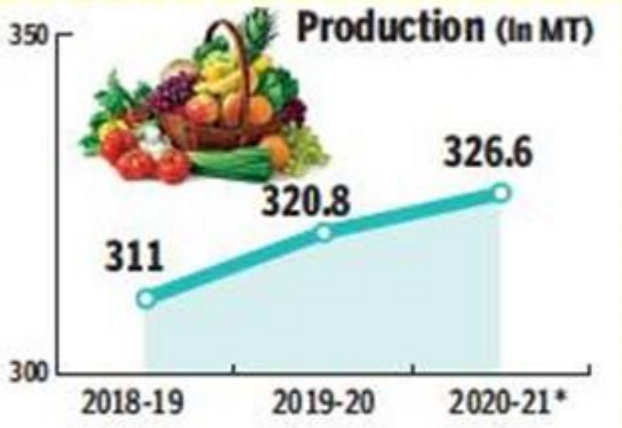
The current level of mechanisation in different types of agricultural activities in cultivation of major crops in India

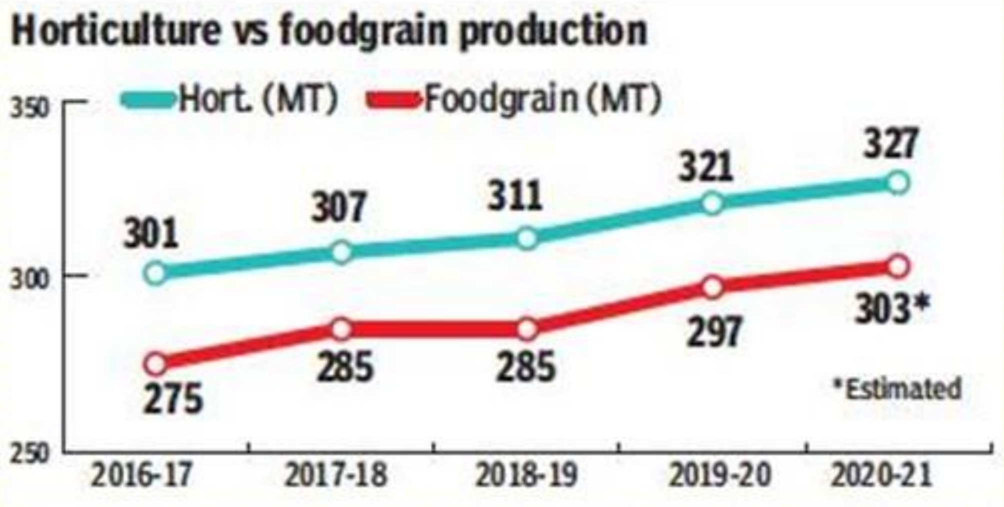
Mechanisation levels for field operations, %

Major crops	Seed bed preparation	Sowing/ planting/ transplanting	Weeding and interculture and plant protection	Harvesting and threshing
Rice	70	20	30	60
Wheat	70	60	50	70
Maize	60	40	30	30
Sorghum and millets	50	30	15	10
Pulses	50	40	20	25
Oilseed	50	40	20	25
Cotton	50	30	25	0
Sugarcane	55	10	20	10

OVERALL HORTICULTURE PRODUCTION IS HIGHER THAN FOODGRAINS PRODUCTION!







BANANA LEADS IN FRUITS PRODUCTION ... POTATO IN VEGETABLES PRODUCTION

(Area in 000 hectares, Production in 000 tonnes)

HORTICULTURE OUTPUT	2019-20 (Final)		2020-21 (Ist Advance Estimates)	
	Area	Production	Area	Production
FRUITS				
Banana	897	32,597	920	33,733
Citrus#	1,075	14,568	1068	14,242
Mango	2,281	20,265	2388	21,124
Total Fruits*	**6,791**	**1,02,028**	**6963**	**1,03,228**
VEGETABLES				
Onion	1,431	26,091	1595	26,292
Potato	2,051	48,562	2247	53,114
Tomato	811	21,173	825	20,148
Total Vegetables*	**10,303**	**1,88,907**	**10711**	**1,93,609**

COCONUT LEADS IN PLANTATION (HORTICULTURE) CROPS ... GARLIC, CHILLIES, TURMERIC ARE THE MAJOR SPICES

HORTICULTURE OUTPUT	2019-20 (Final)		2020-21 (Ist Advance Estimates)	
	Area	Production	Area	Production
PLANTATION CROPS				
Arecanut	745	1,352	743	1,220
Cashewnut	1,125	703	1136	738
Coconut	2,109	13,598	2135	13,787
Total Plantation Crops*	**4,077**	**1,56,799**	**4114**	**15,771**
SPICES				
Chillies (Dried)	684	1,931	743	1,914
Coriander	529	701	552	720
Cumin	1,276	912	1247	887
Garlic	352	2,944	371	3,007
Pepper	258	107	258	123
Turmeric	296	1,153	284	1,106
Total Spices*	**4,351**	**10,298**	**4409**	**10,243**
TOTAL*	**26,457**	**3,20,765**	**27168**	**3,26,578**

20th LIVESTOCK CENSUS SHOWS INCREASE IN SHEEP AND GOATS

S. No.	Species	19th Livestock Census 2012 (no. in millions)	20th Livestock Census 2019(no. in millions)	Growth Rate (%) 2012-19
1	Cattle	190.90	193.46	1.34
2	Buffalo	108.70	109.85	1.06
3	Yaks	0.08	0.06	-25.00
4	Mithuns	0.30	0.39	30.00
	Total Bovines	299.98	303.76	1.26
5	Sheep	65.07	74.26	14.12
6	Goat	135.17	148.88	10.14
7	Pigs	10.29	9.06	-11.95
8	Other animals	1.54	0.80	-48.05
	Total Livestock	512.06	536.76	4.82
9	Poultry	729.21	851.81	16.81

LIVESTOCK CENSUS 2019

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KEY STATISTICS

- The increase of 4.6% in Livestock population is primarily led by sheep and goats. There is marginal increase in cattle and buffaloes.
- The population of cross breed milch cattle was around 26 million and increased by around 32%. However, the indigenous breed milch cattle registered an increase of just less than 1% and stands at around 49 million.

The Department of Animal Husbandry & Dairying, Ministry of Fisheries, Animal Husbandry and Dairying have released the 20th Livestock Census in 2019.

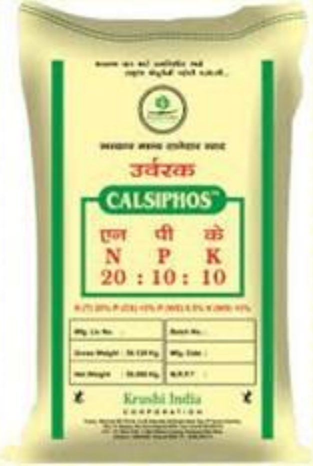
INDIAN FARMER UNDERSTOOD THE ECONOMICS OF ANIMAL REARING!

- Female cattle (cows) increased by 18%, whereas male cattle (oxen) population has seen a decrease of 30% between 2012 and 2019. As a consequence, the male to female cattle ratio dropped to 1:3 from 1:1.8 in the 2012 Livestock Survey.
- Similarly, the population of male buffaloes declined by around 42%. We can conclude that the Indian farmer is strongly influenced by the economics of owning a livestock.

SUBSIDIES AND GREENHOUSE GASES ARE MAJOR CONCERNS IN AGRICULTURE!

- Rice and sugar are heavily subsidised through cheap/free power for irrigation as well as fertilisers, especially Urea.
- Power and fertiliser subsidies account for about 15% of its value in States like Punjab and Haryana. If these subsidies are withdrawn or rationalized through direct income transfers to farmers, rice will not be a preferred crop with the farmers.
- Moreover, for sugar exports, in majority of the years, incentives are given by the Government, to clear excess domestic stocks of sugar. It often leads to a challenge at the WTO by sugar exporting countries like Australia, Brazil and Thailand.
- Rice cultivation contributes to more than 18% of the GHG emissions generated from agriculture.

30% RAISE OF N_2O IN FOUR DECADES GLOBALLY AND THE MAJOR SOURCE IS AGRICULTURE!



- A 2020 review of Nitrous Oxide sources and sinks found that emissions rose 30% in the last 4 decades and N_2O is responsible for roughly 7% of global warming, since pre-industrial times.
- Nitrous Oxide comes from both natural and anthropogenic sources.
- The largest source of Nitrous Oxide is agriculture and majority of the agriculture emissions result from usage of nitrogenous fertilizers and ill-management of animal waste.
- Fossil fuel combustion and industrial processes are the other important sources of Nitrous Oxide emissions.

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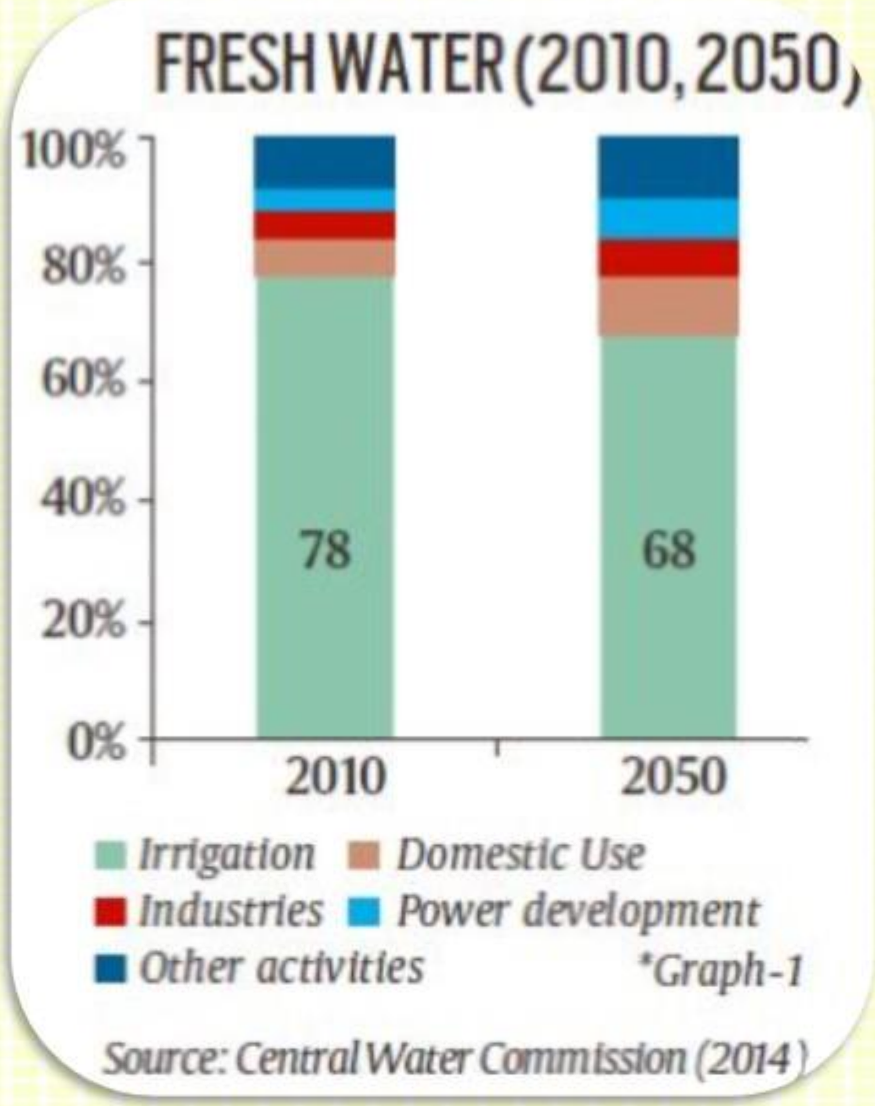
FACTS & FIGURES ABOUT WATER AVAILABILITY IN INDIA



- India has only 4% of the global freshwater resources, whereas, it is home to around 18% of the World population.
- 78% of total water was used for agriculture in 2010, which is likely to be reduced to 68% by 2050.
- For domestic use, it was just 6% in 2010, likely to go up to 9.5% by 2050.

Agriculture will remain the biggest user of water to produce enough food. Water-guzzling crops need to be shifted to geographically sustainable areas.

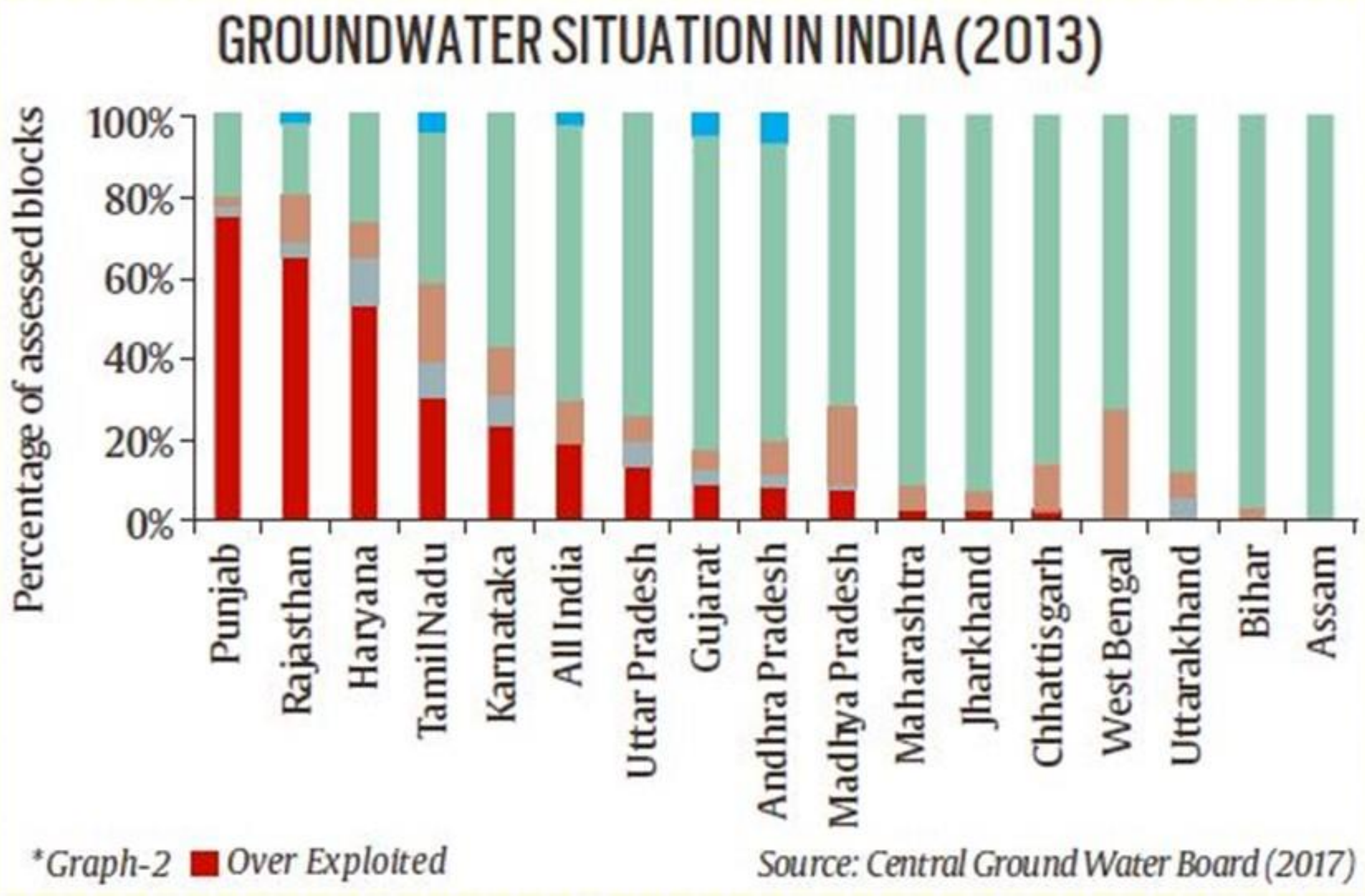
63% IRRIGATION IS THROUGH GROUND WATER!

- Of around 198 million hectares of India's gross cropped area, almost half is irrigated.
- Major sources of this irrigation are groundwater 63%, canals 24%, tanks 2% and all other sources put together is 11%.
- The real burden on irrigating Indian agriculture lies with ground water, driven by private investments from farmers.

VARIOUS STATISTICS ABOUT IRRIGATION IN INDIA

- Agriculture Sector accounts for around 75-80% of total water consumed in the country.
- Right over land gives right over water and this is considered a major flaw, as it leads to unchecked extraction of water.
- At the time of Independence, majority of the irrigation was through canals. However, now the largest component of irrigation is through groundwater, which hovers around 60% of the total irrigated area.
- Out of the total cultivable area, the total irrigated area is hovering in between 40-50%, whereas rain-fed irrigation is still above 50%.

NO EFFECTIVE REGULATION OF GROUND WATER!



- The policy of cheap or free power supply for irrigation led to the situation of near-anarchy in the use of ground water.
- Power subsidies to agriculture cost around Rs.70,000 Cr to Rs. 80,000 Cr. every year to the Govts and are on the increasing trend.
- In places like Punjab, the water table is going down by almost a metre every year and this is going on for the past two decades!

VARIOUS STATISTICS ABOUT AGRICULTURAL CREDIT

- Predominantly, the credit to agriculture sector is disbursed by the public sector banks.
- After liberalization, the share of agricultural credit to the sectoral GDP increased from 20% to around 50%. In fact, during the past five years till 2022, it increased by 10%.
- The biggest problem is that the agricultural credit is highly skewed in favour of crop loans, than long-term loans for land development.
- Another point is that around 10% of the credit goes to dairying and fisheries, whereas the allied sector's share is around 35%-40% in the overall agriculture.

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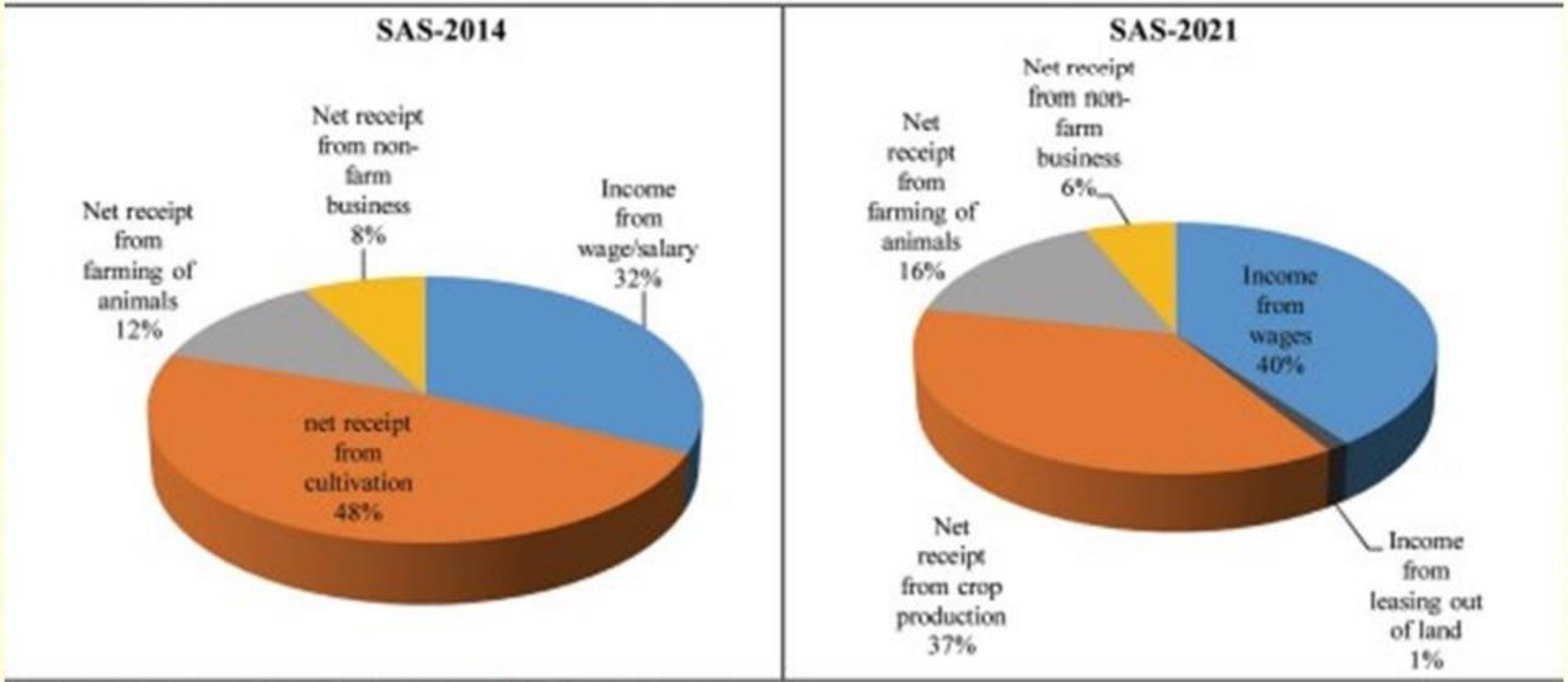
INCOME FROM WAGES OVERTAKES THAT FROM CROP PRODUCTION!

Figure 16: Composition of Average Monthly Income of Agricultural Households



Source: Based on data of SAS, 2014 and SAS, 2021.

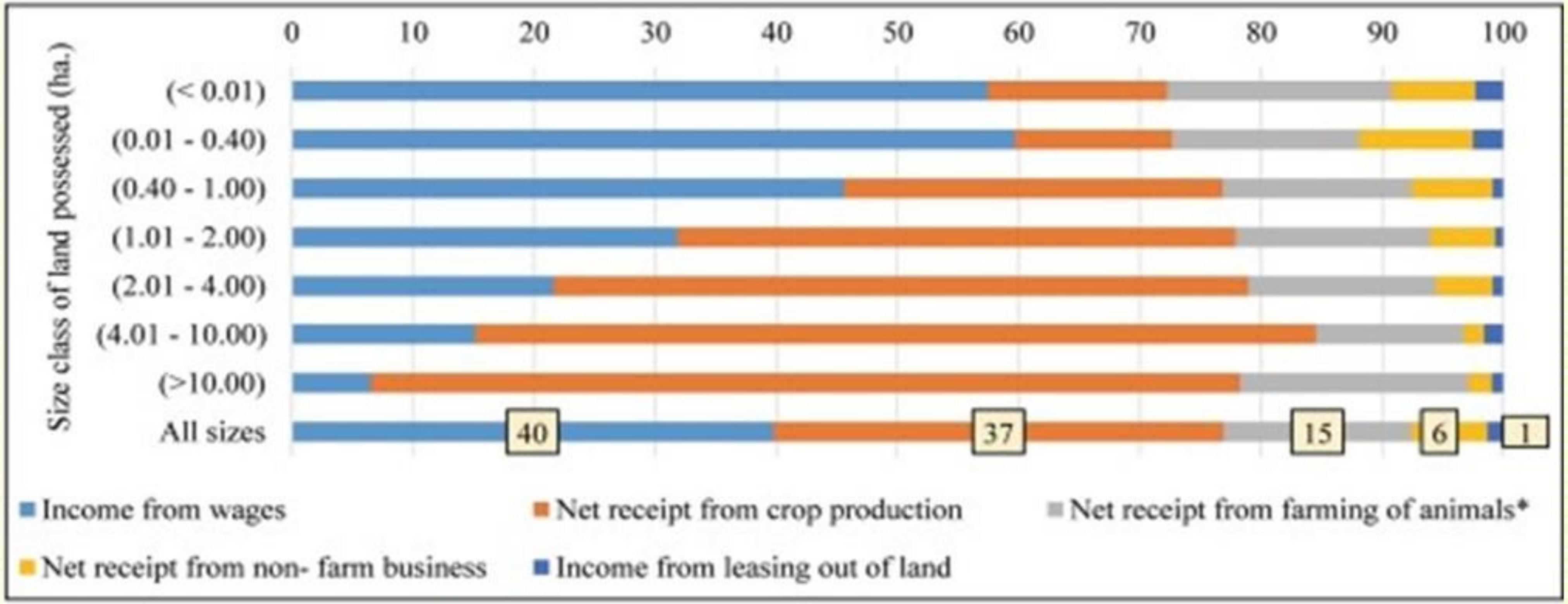
The average monthly income per agricultural household works out to be Rs. 10,218 and as per SAS report of 2014, it was Rs. 6,426.

SITUATION ASSESSMENT SURVEY (SAS)

National Statistical Office (NSO) conducted a survey on “land and livestock holdings of households and the situation assessment of agricultural households” as part of its 77th round in 2019 and the report was released in September 2021.

FOR MARGINAL FARMERS, THE SOURCE OF INCOME IS PRIMARILY FROM WAGES!

Figure 17: Sources of Average Monthly Income Class-wise (per cent)



Source: Based on data of SAS, 2021.

Increasing number of small farmers and increasing importance of livestock requires increased focus on measures like development of small farm technology, promoting non-farm businesses and the development of allied activities, including animal husbandry, dairying and fisheries.

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